Emotion Vector-Based Fine-Tuning of Large Language Models for Age-Aware Teenage Book Recommendations

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Abstract

Reading is a vital skill for teenagers as described by the National Institute of Child Health and Human Development, "Reading is the single most important skill necessary for a happy, productive, and successful life." Yet, teens and their parents often struggle to find engaging books amid an overwhelming number of options. Moreover, existing book recommender systems rely heavily on user data such as profiles, reviews, or browsing behavior—information often restricted for minors due to privacy laws. To address this, we propose a privacy-conscious, teenage book recommender system that analyzes the emotional content of books using the NRC Emotion Intensity Lexicon (NRC-EIL). By extracting emotion vectors from book descriptions, we capture each book's emotional tone and intensity. Our system then uses patterns in emotional preferences across age groups to recommend books that align with teen readers' developmental and emotional needs. While LLMs can make content-based book recommendations for teenagers as well, they still face challenges like training bias, limited sensitivity to age-specific nuances, and lack of transparency. By integrating our emotion vector approach, we fine-tune LLMs to better detect age-relevant emotional cues, enhancing their ability to suggest meaningful and appropriate content for teen audiences. Experimental results confirm that fine-tuning LLMs with our emotional vector approach significantly enhances their ability to generate accurate, age-appropriate book recommendations for teenagers.

CCS Concepts

• **Information systems → Recommender systems**.

Keywords

Emotion vector, age, sentiment analysis, LLMs, teenage books

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1 Introduction

Reading is both an enjoyable pastime and a powerful educational tool. For teenagers, it offers numerous benefits, including improved academic performance, enhanced social awareness, and personal development [12]. However, to fully realize these benefits, it is essential that teens genuinely enjoy the books they read, as enjoyment fosters sustained reading habits and deeper engagement [32]. As such, creating effective book recommender systems for teenagers is vital in encouraging consistent and meaningful reading. Most existing systems, however, rely heavily on a user's past reading history, such as reviews or ratings. Yet this approach poses several challenges: such data is often minimal or difficult to obtain, especially for minors due to privacy laws, and accurately assessing the similarity between books remains a complex task [19, 21]. Moreover, many teen users may not have an established reading profile, making personalized recommendations difficult. To address this, we develop an age-based recommender that leverages observable trends in reading preferences across different age groups. By better understanding the emotional and thematic interests of each age segment, we can support librarians, educators, parents, and teens in discovering books that are both relevant and appealing. Our proposed solution generates an emotion vector to represent the emotional composition of a book, offering a content-based, privacy-conscious foundation for age-aware recommendations.

The emotion vector quantifies the emotional composition of a book by capturing eight primary emotions defined in the National Research Council Canada Emotion Intensity Lexicon (NRC-EIL), which are widely recognized in affective computing literature as fundamental emotional categories [31, 40]. These emotions include *Anger, Anticipation, Disgust, Fear, Joy, Sadness, Surprise*, and *Trust*, in addition to a ninth dimension labeled Objective, representing emotionally neutral or unclassified content. Due to copyright constraints that restrict access to full book texts, we utilize *book descriptions* as the basis for analysis. These summaries are typically accessible and provide sufficient semantic content to infer the dominant emotional themes of a book. Each word or its synonym in a description is mapped to its corresponding emotion intensities using the NRC-EIL. The final emotion vector is computed by aggregating token-level scores from the book description and normalizing to a unit vector. This compact, content-based representation captures the emotional profile and supports downstream tasks.

Although popular, current LLMs are not designed to capture the nuanced emotional composition of teenage books, often overlooking subtle affective patterns that influence engagement. By fine-tuning LLMs with our emotion vector representations, which encode the emotional intensity and distribution derived from book descriptions, we can significantly improve their ability to generate emotionally aligned and age-appropriate recommendations.

The core contributions of the paper are as follows:

(i) We introduce a novel emotion vector representation to capture the emotional distribution/intensity in teen book descriptions;
(ii) We propose a fine-tuning strategy that enhances LLMs' ability to align with age-specific emotional preferences, improving recommendation quality made by LLMs; and
(iii) We construct and evaluate a teenage book recommendation benchmark, demonstrating through empirical results that fine-tuned LLMs outperform both non-fine-tuned LLMs and non-LLM baselines in generating emotionally and developmentally appropriate recommendations.

2 Related Work

Several studies were found related to sentiment analysis specifically applied to books' contents, book reviews, and book descriptions [1, 28]; however, none were found related to teenager's preferences in books in regard to emotion vectors or sentiment analysis. It is well known that children of different ages have different preferences in topics [29], as well as varying tastes in format [13, 14, 24]. However, our approach of analyzing the emotional contents of a book with a focus on teenagers is unique, as the most similar study to ours is [24], which focuses on young children instead, and no other studies construct a similar emotion vector to ours.

Recent advancements in sentiment analysis have significantly expanded its application in recommender systems, particularly for media and literature. A notable line of research focuses on extracting opinion units or aspect-based sentiments from user-generated content such as user reviews enabling fine-grained understanding of user preferences and item characteristics [18, 37, 39]. However, reliance on user reviews presents limitations, especially in domains like children's or teen literature, where such data is sparse.

To overcome this, recent studies have shifted toward content-based emotional modeling, leveraging natural language processing techniques to infer affective information directly from item descriptions, summaries, or transcripts [20, 35]. These approaches often integrate external emotion lexicons, such as the NRC Emotion Intensity Lexicon or VAD-based models, to construct structured emotional profiles of textual content [10]. Additionally, transformer models fine-tuned on emotion-labeled data can recognize complex affective patterns beyond basic sentiment polarity.

In the context of book recommendation, such emotion-driven representations enable more personalized and psychologically aligned suggestions, especially when demographic or behavioral data is limited [2, 15]. Our proposed emotion vector approach, however, builds on this direction by offering a scalable method for quantifying the emotional composition of teenage books using only publicly available descriptions, providing an effective basis for enhancing teenage book recommendation performance of LLMs.

3 Our Teenage Book Recommender

Our age-based and sentiment analysis approach quantifies a book's emotional content by mapping each word in its description to an intensity value from the NRC Emotion Intensity Lexicon and summing these values into a single emotion score. By comparing scores across the descriptions of books favored by teenagers in different age brackets, we reveal how emotional themes shift as readers grow older. This lexicon-based aggregation delivers robust sentiment measurements and highlights clear, age-specific emotional patterns in teenage reading preferences.

Our work is a continuation of Batista [5], taking their work on children's book ratings and expanding the age group to teenagers. We also improve on his work by implementing the python library *synset*, which allows us to check the NRC-EIL with a word's synonyms too, offering us a way to essentially expand the NRC-EIL, lowering the number of words marked as *Objective* and increasing the accuracy of the book's emotion vector. More details follow.

3.1 Emotional Embedding of Teenage Books

To explore the emotional landscape of teenage literature, we applied our emotion vector generation framework to a curated set of books commonly recommended for adolescents, spanning genres such as contemporary fiction, fantasy, and coming-of-age narratives. The goal is to embed each book into an emotion vector space that reflects its emotional signature—anchored in the NRC-EIL's nine dimensions—and to examine how these vectors can capture emotional themes that resonate with teenage readers.

3.1.1 Token-Level Emotion Scoring. Each book description is tokenized, and for each token, we extract emotion intensity scores across the eight NRC-EIL categories plus the Objective dimension. For instance, the sentence "A heart-wrenching story of loss, hope, and unexpected friendship" yields high scores in *Sadness, Joy*, and *Trust*, with lower *Anger* or *Disgust* values.

3.1.2 Vector Aggregation and Normalization. The individual token scores are aggregated by summing across each emotion dimension and then normalized to produce a unit-length vector, i.e.,

$$\vec{e} = \frac{1}{\| \sum_{i=1}^{n} \vec{w_i} \|} \sum_{i=1}^{n} \vec{w}_i \tag{1}$$

where $\vec{w}_i$ is the emotion score vector for token i, and n is the number of tokens in a description d. The result is a 9-dimensional vector $\vec{e}$ representing the book's emotional composition through d.

3.1.3 The Novelty of Our Vector Embeddings. Teenagers experience heighten emotional variability and intensity. Our emotion vectors preserve the *emotional contours* of books that may help teens feel seen, understood, or guided through personal challenges. These vectors are compact and interpretable, using a 9-dimensional emotion vector to distill rich narrative affect into a compact representation that can be used for clustering, similarity search, or emotional trajectory modeling, without requiring full text access. Moreover, emotional embeddings can be aligned with age-specific emotional needs. For instance, early teens may seek reassurance, i.e., *Trust* and *Joy*, while older teens may gravitate toward complex emotional experiences, i.e., *Fear, Sadness*, and *Anticipation*, enabling emotionally congruent book matching. Furthermore, our Emotion vectors enable fine-grained operations such as filtering books based on emotional needs, e.g., low *Fear* for anxious readers, visualizing emotional diversity in a book collection using dimensionality reduction, and detecting emotional gaps.

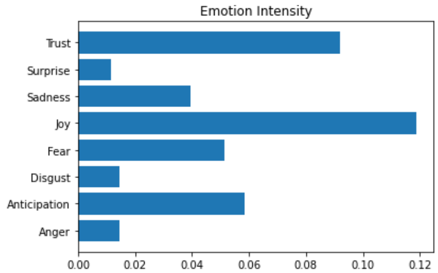

Figure 1: *Emotion vector* of *The Notebook,* Nicholas Sparks

3.2 Generation of Emotion Vectors from Text

To conduct semantic analysis for each book, we compute an emotion vector based on its description. This vector is a normalized representation that captures the relative intensities of eight core emotions defined by the NRC Emotion Intensity Lexicon (NRC-EIL) as presented in Section 1. Our findings indicate that book descriptions—readily available from sources such as GooodReads [33]—serve as an effective proxy for full texts, offering substantial semantic insight while significantly reducing computational cost.

The process for generating an emotion vector of a book involves several steps: first, the description is preprocessed by removing *stop words*. Hereafter, for each remaining word, *emotional intensity* values are retrieved from the NRC-EIL. These values are aggregated across the entire description and normalized to produce a unit vector representing the emotional composition of the book. (Consider the emotion vector of *The Notebook* by Nicholas Sparks, as shown in Figure 1. The book, which has an average rating of 4.1 out of 5.0 and an average reader age of 16.8 extracted from Goodreads, presents a deeply emotional and poignant narrative.)

3.3 Emotion Modeling Using NRC-EIL

The NRC-EIL provides a comprehensive list of English words annotated with intensity scores (ranging from 0 to 1) for eight fundamental emotions: *Anger, Anticipation, Disgust, Fear, Joy, Sadness, Surprise*, and *Trust*. To ensure that variations in the length of book descriptions do not influence the resulting emotion vectors, we normalize each vector to unit length. This normalization enables meaningful comparisons by focusing on the relative proportions of emotions, i.e., its direction, rather than its magnitude [6, 22]. According to Plutchik [26], these emotions form opposites. For example, *Anger* and *fear* are opposites in the sense that one implies *attack* and the other *fight*, and *Joy* and *sadness* are opposites in the sense that one implies *possession* or *gain* while the other implies *loss*. In addition, words with the *highest intensity* scores in the NRC-EIL most effectively represent the *core* emotional dimensions. For instance, "happiest" is the top-scoring word for "Joy", and "trustfulness" for "Trust" [11, 25]. Each word has a defined value in each emotion, which by default is zero, and all of our vectors get normalized. Normalization stops longer descriptions from inflating emotion scores, ensuring fair comparisons across text lengths.

3.4 Synonym Matching for Emotion Mapping

To enhance recommendation quality in content-based systems, we refined the construction of emotion vectors by reducing the frequency of *Objective* labels—default assignments when terms are absent from the NRC-EIL [11]. Leveraging the Python *synset* library, which accesses WordNet [23], we expanded coverage by retrieving *synonyms* for each word and querying their emotional scores. This method—limited to the top two semantic senses to preserve relevance—significantly improves emotion vector granularity and better captures affective nuances essential for emotion-aware recommendations [4, 8]. Although slightly more computationally expensive, this synonym-based augmentation leads to measurable gains in emotional representation, crucial for aligning book recommendations with age-sensitive reader preferences.

There is a statistically significant difference (highlighted) between creating the vectors *without* synset and *with* synset for almost every emotion ($p < 0.01$), as shown below in Table 1.

Table 1: Statistical significant differences (highlighted cells) between vectors calculated *with* and *without* synset

Anger	**0.00063**	Antic.	**0.00018**	Disgust	**0.00276**
Fear	**0.00499**	Joy	0.09231	Sadness	**0.01133**
Surprise	0.18290	Trust	**0.00020**	Objective	**0.00001**

3.5 Modeling Reader Preference via Emotions

Among the eight primary emotions, *Fear, Sadness*, and *Anger* exhibit a consistent age-dependent trajectory: minimal representation in books favored by 12-13-year-olds, a pronounced peak among 14-15-year-olds, and a subsequent decline in older adolescents. This progression suggests early adolescents' avoidance of *intense negative* affect, growing engagement in mid-teens, and eventual preference for more nuanced narratives as older teens regard sensational content as lacking depth. Conversely, *Joy* shows an inverse profile—highest among the youngest cohort and decreasing with age—while *Surprise* also peaks at ages 12-13, highlighting early adolescents' affinity for uplifting and unexpected story elements.

We also find significant differences between books of different age groups (see Table 2 for details). Pairwise significance tests ($p < 0.05$) are performed on each emotion dimension of the normalized emotion vectors, and the significant entries are **highlighted**. A highlighted cell at the intersection of emotion E (row) and age cohorts A_1 vs. A_2 (columns) indicates that, for two titles whose emotion vectors differ only in dimension E, the mean ratings given by cohorts A_1 and A_2 diverge significantly. This table of emotion-age interactions offers concrete guidance for calibrating emotion-aware recommendation models to the nuanced affective preferences of distinct teenage age groups.

3.6 LLMs for Teenage Book Recommendations

LLMs provide a flexible, content-driven basis for book recommendation by encoding nuanced semantic relationships from descriptions, reviews, and metadata [7, 9]. However, as general-purpose language models rather than dedicated recommender engines, LLMs must be fine-tuned on domain-specific interaction data or adapted with recommendation-oriented objectives to reach optimal performance [30, 36]. Through fine-tuning with our designed emotion and age-specific vectors, we can significantly improve LLMs' ability to capture affective patterns and developmental preferences critical for teenage book recommendations. This method adds transparency and contextual alignment to LLMs, extending beyond the generic capabilities of pre-trained LLMs [16, 17, 38]. We enhance the transparency of LLM-based teenage book recommendations by providing an interpretable and content-grounded

Table 2: Statistically significant differences between *very good* books for each age group comparison, separated by emotion. Highlighted cells indicate statistical significance ($p < 0.05$) based on the Wilcoxon Signed-Rank test

Emotion	12-13 vs 14-15	12-13 vs 16-17	12-13 vs 18-19	14-15 vs 16-17	14-15 vs 18-19	16-17 vs 18-19
Anger	**0.0293**	0.1125	0.23140	0.22967	0.24195	0.83791
Anticipation	0.3695	0.7946	**0.01216**	0.32776	**0.00001**	**0.00001**
Disgust	0.09787	0.2462	0.37345	0.33101	0.39000	0.89387
Fear	0.05266	0.2837	0.82511	0.16021	**0.01637**	0.10308
Joy	0.17690	0.1473	**0.00977**	0.88700	0.12560	**0.03917**
Sadness	**0.0135**	**0.0203**	0.18561	0.58271	0.21025	0.3245
Surprise	0.76770	0.1383	0.05581	**0.00402**	**0.00021**	0.09584
Trust	0.42280	0.7976	0.08659	0.45831	0.27306	0.07104
Objective	0.07983	0.7019	**0.02296**	0.12980	**0.00088**	**0.01031**

rationale for why certain books are suggested. Our system allows LLMs to explain what emotional needs a book satisfies, and above all it considerably improves the transparency of LLMs.

3.7 LLM Refinement with Emotion Vectors

Fine-tuning enables targeted performance improvements on specific tasks while avoiding the prohibitive computational demands of training LLMs from scratch, making the approach both *efficient* and *scalable*. In our framework, fine-tuning is applied to adapt LLMs for generating relevant and engaging book recommendations specifically tailored to teenage readers.

To fine-tune an LLM for teenage book recommendations, we incorporate age-specific emotion vectors (constructed from the NRC-EIL) into the training process. Each input pairs a *book description* with its corresponding normalized *emotion vector* and *target age group*, enabling the model to learn patterns that align with emotional and developmental preferences. These features guide the LLM during fine-tuning to generate more emotionally attuned and age-appropriate book suggestions. We evaluate the performance of the fine-tuned models—LLaMA, Qwen, and Gemma—on a book recommendation dataset using standard metrics, i.e., Precision@K (P@K), Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG). Figure 2 illustrates the fine-tuning process of an LLM, which utilizes emotion vectors generated by our model combined with book descriptions.

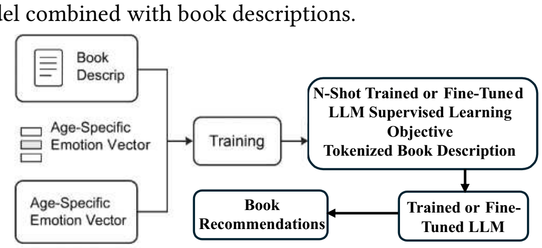

Figure 2: The fine-tuning process of LLMs

3.8 The Role of LLMs in the Integrated system

In our framework, LLMs are not merely content processors—they are emotionally attuned, age-sensitive decision-makers that leverage deep linguistic understanding to recommend books that speak to the unique emotional realities of adolescence. Through emotion-vector fine-tuning, we extend the general-purpose capabilities of LLMs into a specialized, transparent, and developmentally informed recommendation engine tailored to teenage readers.

By integrating our interpretable emotion vectors into the LLM pipeline, this hybrid representation—semantic knowledge enriched with structured emotion and age conditioning—enables the LLM to perform personalized, context-sensitive recommendations. Moreover, it supports explanatory outputs that justify selections based on affective alignment, improving system transparency and interpretability. For example, the model may recommend a book because its emotional vector reflects themes of "Sadness" and "Trust" that are commonly favored by older teens undergoing emotional maturation. Through this design, LLMs in our system function as more than ranking engines; they become emotionally attuned agents capable of guiding teenage readers toward books that match both their current affective state and developmental needs. This represents a substantial advance over existing recommender systems and underscores the value of fine-tuning LLMs with task-specific emotional and cognitive constraints.

4 Performance Analysis

We evaluate the impact of fine-tuning LLMs with the proposed emotional vectors through a comprehensive analysis using standard quantitative metrics based on emotional alignment scores. These scores are determined by measuring the degree of similarity between the emotional preference expressed by a reader identified by his age and the emotional tone inferred from recommended books, measured via cosine similarity between their respective emotional vector representations. The source code and results are available at https://github.com/Epic-Doughnut/LLM_Semantics.

4.1 Dataset for Performance Evaluation

The dataset used for evaluating the performance of our teenage book recommender is the *Goodreads* dataset. The dataset, which is a collection of data posted on the Goodreads website (goodreads.com), primarily focusing on information about books, their authors, and user interactions. It includes book metadata, user ratings and reviews, and other details like genres and publication dates. It contains 2.3 million books, 0.8 million users, and 228 million user-book interactions, and 104 million ratings. The dataset is a robust benchmark for evaluating book recommender systems, with unparalleled scale, diverse user preferences, rich rating data, and metadata—including user ratings and ranking-based (teenage) book recommendations—that serve as ground truth for any performance evaluation.

4.2 An Empirical Study on Enhancing LLMs

In this section, we address the following research questions (RQs):

Table 3: LLM's fine-tuning performance metrics where bold-faced numbers indicate improvement

Model	Fine-Tuned	P@3	P@5	MAP	MRR@1	MRR@5	nDCG@3	nDCG@5
LLaMA 3.1 8B	No	0.633	0.560	0.611	0.683	0.683	0.695	0.707
	Yes	0.633	**0.680**	**0.689**	**0.832**	**0.871**	**0.774**	**0.823**
Qwen 2.5 7B	No	0.833	0.780	0.868	0.940	0.940	0.950	0.975
	Yes	**0.967**	**0.900**	**0.943**	0.940	0.940	**0.972**	**0.976**
Gemma 3 4B	No	0.600	0.640	0.681	0.817	0.800	0.777	0.757
	Yes	**0.633**	**0.671**	0.612	0.814	**0.870**	0.663	**0.792**

- **RQ1**. What performance improvements are observed when comparing fine-tuned LLMs to their non-fine-tuned counterparts?
- **RQ2**. What are the intrinsic limitations of applying the emotional vector approach to fine-tune LLMs for enhancing recommendation accuracy in teen-focused book selection?
- **RQ3**. What do performance gains and regressions reveal about the robustness of fine-tuning with emotional vector integration and its optimization potential for LLMs?

We evaluated the effectiveness of emotional vector integration by fine-tuning an LLM on books recommended for readers aged 12-19, measuring its ability to generate emotionally aligned suggestions for a given teen book, ranked by their relevance.

The empirical study was run independently on three LLMs: LLaMA 3.1 8B, Qwen 2.5 7B, and Gemma 3 4B. LLaMA 3.1 8B is designed for optimizing efficiency, Qwen 2.5 7B for excelling in context-aware understanding, and Gemma 3 4B for targeting multilingual tasks. We compared each model's performance before and after fine-tuning using various metrics, as shown in Table 3.

We address each of the research questions, RQ1, RQ2, and RQ3, using the results shown in Table 3 from our empirical study. The findings displayed in the table are statistically significant ($p < 0.05$).

RQ1: Table 3 highlights the consistent enhancement of LLMs performance in teen book recommendations through fine-tuning, especially in precision and contextual alignment, though the degree of improvement varies across models. Notably, LLaMA 3.1 8B shows clear improvements across all metrics after fine-tuning except P@3, which remains unchanged. In comparison, Qwen 2.5 7B shows the most significant boost, with all metrics improving—especially P@3—despite MRR@1 and MRR@5 remaining the same after fine-tuning. While fine-tuning Gemma 3 4B does not yield substantial overall changes, the top-5 ranked values consistently improve. The fine-tuned Gemma 3 4B underperforms, likely due to its design for multilingual rather than domain-specific tasks.

RQ2: A key challenge in applying the emotional vector approach to teenage book recommendations is the model's limited ability to generalize across diverse emotional responses, especially when emotional needs shift rapidly due to social, academic, or contextual factors. While our age-based approach captures generalized emotional patterns, such fluctuations remain difficult to model consistently. LLM performance is further influenced by model scale and fine-tuning data size, with larger models requiring substantial, representative datasets to capture nuanced emotional and contextual patterns—exemplified by LLaMA 3.1 8B, which is significantly larger than Gemma 3 4B as shown in Table 3.

RQ3. We observe that while the fine-tuning strategy proves effective for certain models, further investigation is required to understand its limitations and optimize it for LLMs like Gemma 3 4B.

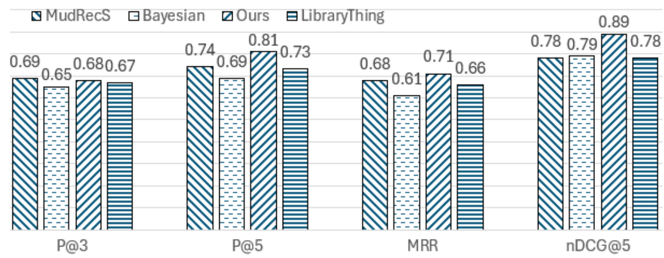

Figure 3: Assessing recommenders for teenage books

Key factors, such as model size, internal setups, and training strategies, must also be carefully considered. Further optimization that includes incorporate cross-modal data (e.g., text, voice, or visual sentiment) to enhance emotional vectors by capturing a broader spectrum of emotional cues can be applied in future design.

4.3 Evaluating Fine-Tuned LLM Recommender

In this section, we present existing book recommenders to be compared with our own. These recommenders were chosen, since they achieve high accuracy in recommendations on books based on their respective model and are emerging book recommender systems.

MudRecS [27] recommends books based on content similarity to those a user likes, without relying on access history, social connections, collaborative filtering, or personal traits like gender or age. Using Bayesian analysis of survey data from 11-15-year-old students (grades 6-9), Vuong et al. [34] found that reading interest is linked to parental reading habits, genre preferences, and advanced recommenders that promote reading. LibraryThing [3], a platform for book cataloging and discovery, allows users to search for books tagged as 'Young Adult.' Figure 3 shows performance results for four book recommenders, including *ours*—the fine-tuned LLaMA 3.1 8B—highlighting its superior, statistically significant performance ($p < 0.035$).

5 Conclusion

Teen reading is vital for cognitive and emotional development, yet many adolescents struggle with engagement due to the constant pull of social media and streaming platforms and short-form digital content that compete for their attention. The proposed book recommender systems matches teenagers' interests and emotional needs, making reading more engaging and accessible. Our age-based, emotional-vector approach outperforms existing methods by effectively aligning with teens' emotional and developmental profiles. Experimental results support the design objective and show that fine-tuning LLMs improves their performance in offering teen book recommendations.

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